Noah Wilson

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Robotics • Mechatronics • Dexterous Manipulation • Robot Learning

EDUCATION

University of Florida, Gainesville, FL
B.S. Mechanical Engineering, Honors Program — Expected May 2028
GPA: 3.66/4.00 | Dean's List: Fall 2025, Spring 2026

TECHNICAL SKILLS

CAD & Manufacturing: SolidWorks (CSWA), Fusion 360, Onshape, GD&T, Drawings, FEA, Additive Manufacturing
Programming & Software: Python, C/C++, ROS 2, PyTorch, Linux, Git, NumPy, Pandas, Java
Robot Learning: Reinforcement Learning, Physics Simulation, MuJoCo Playground, JAX, Brax, PPO, AWS EC2
Hardware: Sensor Integration, Tactile Sensing, DYNAMIXEL, ESP32, STM32, Embedded Systems, Soldering

EXPERIENCE

AeroVironment | Mechanical Engineering Intern — Sunrise, FL | Jun 2026 – Aug 2026

- Redesigned a production electronics test fixture using tolerance stack-up analysis and a custom 3D-printed base, integrating power delivery, PCB mounting, hardware, and computer interfaces
- Developed a production alignment fixture through CAD, SolidWorks FEA, prototyping, and testing; refined tolerances and engineering drawings with manufacturing technicians
- Created a BOM covering 120+ mechanical and electrical components, verified compatibility against datasheets, and managed part numbers, vendors, and documentation in PLM/PDM

UF ICIC Lab | Undergraduate Researcher - Dexterous Robotics — Gainesville, FL | Apr 2026 – Present

- Redesigned and integrated a 12-DOF hand exoskeleton with tactile sensors, cameras, encoders, and embedded electronics for dexterous manipulation data collection
- Conducted a 20-participant study across four manipulation tasks, collecting synchronized tactile, visual, wrist-pose, and encoder data from paired bare-hand and exoskeleton demonstrations
- Operated the data-acquisition system and diagnosed sensor and electromechanical faults through recalibration, soldering, and hardware repair
- Coauthored two manuscripts submitted to IEEE ICRA 2027
- Created and independently presented a tactile-sensing and edge-AI poster to ONR leadership

UF Bio-Sensing, Automation & Intelligence Lab | Undergraduate Researcher — Gainesville, FL | Aug 2026 – Present

- Designed and iterated a blueberry-harvesting end effector with synchronized geared linkages, compliant dual-bowl grippers, and a ROBOTIS DYNAMIXEL servo
- Integrated mechanical and electrical hardware and wrote Python control scripts, configuring travel limits, speed, torque, and recovery procedures
- Diagnosed binding and insufficient torque, redesigned and FDM-printed ASA components, and validated operation over 10 consecutive cycles

Gator Motorsports | Drivetrain Subteam — Gainesville, FL | Aug 2025 – Present

- Support Formula SAE drivetrain design, integration, and fabrication through physics-based analysis

PROJECTS

Wearable Robotic Third Thumb | Independent Project — Jul 2026 – Present

- Designed and integrated a tendon-driven wearable robotic thumb using custom 3D-printed components, dual servos, an ESP32, Bowden-cable actuation, and soldered power circuitry
- Tested thumb closure under load using load cells, multimeters, and bench power supplies; identified limited lateral motion and initiated a second-generation design

Humanoid Reinforcement Learning | Independent Project — May 2026 – Aug 2026

- Built a cloud-based robot-learning pipeline using AWS EC2, MuJoCo Playground, JAX, and Brax PPO, with experiment logging for training and evaluation
- Trained a Unitree G1 locomotion policy for 202M+ simulated transitions and evaluated its behavior in simulation
- Evaluated 16 G1 and Panda environments across reward shaping, curricula, resets, and reference tracking